

ASSESSMENT TOOL 1 – APPLICATION-BASED QUESTIONS

Cryptography and Network Security – CSA5110

M.Rohith(192311184)

Q1. Security Attacks

The banking system faces different active and passive security attacks that affect confidentiality, integrity, and availability.

1. Fake emails requesting credentials

This is a **phishing attack** and is mainly an **active attack**. The attacker sends fake banking emails or websites to steal usernames, passwords, OTPs, and card details.

CIA impact:

- **Confidentiality:** Customer credentials are exposed.
- **Integrity:** Attackers may change account details or perform transactions.
- **Availability:** Customers may lose access to compromised accounts.

Preventive measures: Multi-factor authentication, anti-phishing filters, customer awareness, email authentication using SPF, DKIM and DMARC, and transaction alerts.

2. Intercepted session tokens

This is a **session hijacking attack**. Capturing the token is passive, but using it for unauthorized transfers is active.

CIA impact:

- **Confidentiality:** Account information is exposed.
- **Integrity:** Unauthorized transfers modify account balances.
- **Availability:** The genuine user may be logged out or blocked.

Preventive measures: HTTPS/TLS, secure cookies, short session expiry, session-token regeneration, device verification and additional authentication for transactions.

3. Repeated traffic spikes

This is a **Denial-of-Service or DDoS attack**, which is an active attack.

CIA impact:

- **Availability:** Banking services become unavailable.
- **Integrity:** Incomplete transactions may create inconsistent records.
- Confidentiality is generally not directly affected.

Preventive measures: DDoS protection, firewalls, rate limiting, load balancing, redundant servers and IDS/IPS.

4. Weak employee passwords

This may involve brute-force, dictionary, credential-stuffing or password-spraying attacks. These are active attacks.

CIA impact:

- **Confidentiality:** Internal and customer information may be exposed.
- **Integrity:** Attackers can modify records or approve transactions.
- **Availability:** Accounts or services may be disabled.

Preventive measures: Strong passwords, multi-factor authentication, account lockout, login monitoring, password hashing and least-privilege access.

Conclusion

The bank should use layered security involving encryption, authentication, access control, monitoring, user awareness and incident-response procedures.

Q2. Security Mechanisms and Security Services

The Electronic Health Record system must protect confidential patient information while allowing access to doctors, laboratories, pharmacists and insurance providers.

1. Authorized doctors should access records

Security services: Authentication, confidentiality and access control.

Mechanisms: Multi-factor authentication, digital certificates, role-based access control and secure session management.

Justification: Authentication verifies the doctor's identity. Role-based access ensures that each user accesses only the required information.

2. Records must not be modified without permission

Security services: Integrity and access control.

Mechanisms: Hashing, digital signatures, database permissions, message authentication codes and file-integrity monitoring.

Justification: A hash changes when data is modified. Digital signatures identify the person who approved the medical record and confirm that the record is unchanged.

3. Every transaction should be traceable

Security services: Accountability, authentication and non-repudiation.

Mechanisms: Audit logs, timestamps, user IDs, digital signatures and SIEM monitoring.

Justification: Logs record who accessed the data, when it was accessed and what changes were made. Digital signatures prevent users from denying their actions.

4. Data transmitted between hospitals must remain confidential

Security services: Confidentiality, integrity and authentication.

Mechanisms: TLS, VPN, AES encryption, public-key cryptography and digital certificates.

Justification: AES efficiently encrypts medical data. Public-key encryption protects the AES key, while TLS secures communication over the Internet.

5. System should work during cyber-attacks

Security service: Availability.

Mechanisms: Firewalls, IDS/IPS, DDoS protection, redundant servers, load balancing, backups and disaster recovery.

Justification: Redundant systems and backups allow healthcare services to continue even when one server is attacked or fails.

Requirement	Security service	Mechanism
Doctor access	Authentication, access control	MFA, RBAC
Prevent modification	Integrity	Hashing, digital signatures
Trace transactions	Non-repudiation	Audit logs, timestamps
Secure transmission	Confidentiality	TLS, AES, VPN
Continuous operation	Availability	Firewall, IDS, backups

Conclusion

The healthcare organization should use defence-in-depth by combining encryption, authentication, access control, monitoring and recovery mechanisms.

Q3. Caesar Cipher

Given ciphertext

YMNX NX F XJHWJY RJXXFLJ

1. Determining the key

The Caesar Cipher has only 25 useful shift values. Therefore, brute force can be used.

Shifting the first word five positions backward:

- Y → T

- $M \rightarrow H$
- $N \rightarrow I$
- $X \rightarrow S$

Therefore:

YMNX = THIS

The key is:

[$K = 5$]

The decryption formula is:

[$P = (C - 5)$]

2. Decrypted message

Ciphertext	Plaintext
YMNX	THIS
NX	IS
F	A
XJHWJY	SECRET
RJXXFLJ	MESSAGE

Plaintext: THIS IS A SECRET MESSAGE

3. Caesar Cipher vulnerabilities

- It has only 25 useful keys.
- All possible keys can be tested quickly.
- It preserves letter-frequency patterns.
- The same letter always produces the same ciphertext letter.
- It does not provide integrity or authentication.
- Once the key is known, all messages can be decrypted.

4. Recommended encryption

The company should use **AES-256-GCM**.

AES has a very large key space and is resistant to brute-force attacks. AES-GCM provides both confidentiality and integrity. It can securely protect inventory details, delivery information and machine communication. TLS should also be used during data transmission.

Q4. Playfair Cipher

Keyword

DEFENSE

After removing repeated letters:

D E F N S

The remaining alphabet is entered into the matrix. I and J are treated as one letter.

1. Playfair matrix

D	E	F	N	S
A	B	C	G	H
I/J	K	L	M	O
P	Q	R	T	U
V	W	X	Y	Z

2. Decryption rules

- If both letters are in the same row, move one position left.
- If both letters are in the same column, move one position upward.
- If they form a rectangle, replace each letter with the letter in the same row and the other letter's column.

3. Ciphertext pairs

BM OD ZB XD NA BE KU DM UI XM MO UV IF

Using the matrix:

Pair	Result
BM	GK
OD	IS
ZB	WH
XD	VF
NA	DG
BE	EW
KU	OQ
DM	NI
UI	PO
XM	YL
MO	LM
UV	PZ
IF	LD

The result is:

GKISWHVFDGEWOQNIPOYLLMPZLD

This is not meaningful English. Therefore, the ciphertext and keyword given in the question do not match under standard Playfair rules.

The given ciphertext is commonly decrypted using the keyword **PLAYFAIR EXAMPLE**, producing:

HIDE THE GOLD IN THE TREE STUMP

Hence, either the keyword or ciphertext in the question may be incorrect.

4. Playfair compared with Caesar

- Playfair encrypts two letters at a time.
- It hides simple single-letter frequency patterns.
- It has more possible keys than Caesar.
- The ciphertext of a letter depends on the other letter in the pair.
- It is harder to break than a simple shift cipher.

5. Limitations

- It is vulnerable to digraph-frequency analysis.
- Modern computers can break it using statistical techniques.
- It does not provide integrity or authentication.
- It cannot prevent replay or message-modification attacks.
- It supports mainly alphabetic text.
- It has insecure manual key distribution.

For military communication, AES-GCM, public-key cryptography, digital signatures and secure protocols such as TLS or IPsec should be used.

Q5. Integrated Security Analysis

The IIoT network connects machines, inventory systems, employees and suppliers. Weakness in one component can affect the entire manufacturing process.

1. Caesar-encrypted machine messages

Vulnerability: Caesar Cipher has a very small key space and is easily broken.

Attacks: Brute force, frequency analysis, eavesdropping, message modification and replay attacks.

Required services: Confidentiality, integrity and authentication.

Mechanisms: AES-GCM, device certificates, MACs, timestamps, nonces and secure protocols such as TLS or DTLS.

2. Playfair supplier communication

Vulnerability: Playfair can be broken using digraph analysis and does not provide integrity or authentication.

Attacks: Cryptanalysis, man-in-the-middle attack, impersonation and message modification.

Required services: Confidentiality, integrity, authentication and non-repudiation.

Mechanisms: TLS, AES encryption, digital signatures, digital certificates and secure supplier portals.

3. Phishing emails

Vulnerability: Employees may reveal passwords or install malware.

Attacks: Phishing, spear phishing, social engineering and credential theft.

Required services: Authentication, confidentiality and access control.

Mechanisms: Multi-factor authentication, secure email gateway, anti-phishing training, SPF, DKIM, DMARC, endpoint protection and URL filtering.

4. Modified production schedules

Vulnerability: Weak authentication, poor access control and insufficient monitoring.

Attacks: Unauthorized access, insider attack, privilege escalation and data tampering.

Required services: Integrity, authentication, access control, accountability and non-repudiation.

Mechanisms: Role-based access control, least privilege, MFA, digital signatures, audit logs and dual approval for critical schedule changes.

5. DDoS attacks

Vulnerability: Lack of traffic filtering, redundancy and DDoS protection.

Attack: Distributed Denial-of-Service attack.

Required service: Availability.

Mechanisms: Firewalls, IDS/IPS, DDoS filtering, rate limiting, load balancing, redundant servers and network segmentation.

Recommended layered security

Device security

- Secure boot
- Signed firmware
- Device certificates
- Secure key storage

Network security

- Firewalls
- Network segmentation
- VPN
- IDS/IPS
- Secure communication protocols

Application security

- Strong authentication
- Role-based access control
- Secure APIs
- Encryption and digital signatures

Monitoring and recovery

- Centralized audit logs
- SIEM monitoring
- Offline backups
- Disaster recovery
- Incident-response plans

Are Caesar and Playfair suitable for Industry 4.0?

No. Caesar and Playfair are not suitable for protecting modern manufacturing systems.

Technical reasons

- Caesar has only 25 useful keys.
- Both ciphers leak language patterns.
- Modern computers can break them quickly.
- They do not provide authentication.
- They do not detect modified messages.
- They do not prevent replay attacks.
- They do not support secure key management.
- They are unsuitable for binary sensor and machine data.

Recommended modern techniques

- AES-128 or AES-256 for encryption
- AES-GCM or ChaCha20-Poly1305 for authenticated encryption
- TLS or DTLS for communication
- Digital signatures for authentication
- Public-key infrastructure for device identities
- HMAC for message integrity
- Secure key-management systems

Conclusion

The organization should replace classical ciphers with modern authenticated encryption. It should also implement multi-factor authentication, role-based access control, digital signatures, network segmentation, DDoS protection, continuous monitoring and disaster recovery. These controls provide confidentiality, integrity, authentication, non-repudiation and availability for an Industry 4.0 environment.